■ LOCAL AREA NETWORK DESIGN PROJECT

1. Introduction

This report presents the design and simulation of a Local Area Network (LAN) for a residential home. The LAN includes five computers, a wireless access point, and a network printer. Cisco Packet Tracer was used to model and test the network.

2. Network Components

Router: Provides gateway access, assigns IPs (DHCP), and routes traffic.
Switch (2960): Connects all wired devices on the LAN.
Wireless Access Point: Enables Wi-Fi connectivity for wireless clients.
Desktop Computers (5): Connected via Ethernet or Wi-Fi.
Network Printer: Assigned a static IP for easy identification.
Ethernet Cables: Straight-through cables used for all wired connections.

3. Network Topology

The router connects to the switch, which links all five computers, the printer, and the wireless access point. The wireless access point enables mobile devices to join the LAN.

4. IP Addressing Scheme

Network: 192.168.10.0/24
Router: 192.168.10.1
WAP: 192.168.10.20
Printer: 192.168.10.50
DHCP Range: 192.168.10.100 – 192.168.10.150

5. Device Configuration Summary

Router: Configured with IP 192.168.10.1, DHCP enabled, DNS set to 8.8.8.8.
Switch: Default VLAN 1; used for wired communication.
WAP: SSID = HomeLAN, Security = WPA2-PSK, Password = MyHome123.
Printer: Static IP = 192.168.10.50.

6. Testing & Verification

All PCs received DHCP IP addresses successfully. Ping tests between devices were successful, confirming full network connectivity. Wireless connectivity was verified via the access point, and the printer responded to network pings.

7. Group Members & Roles

Member 1: Network Design & Topology
Member 2: Device Configuration
Member 3: Documentation & Final Report

8. Conclusion

The designed LAN meets all required criteria, supports both wired and wireless communication, and functions reliably as tested in Cisco Packet Tracer.